

Solution Requirements

Date	01 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>No authentication required; users can directly access and use the HDI Predictor.</i>	Low
2	Authorization levels	<i>No role-based access required; all users can access prediction and project information.</i>	Low
3	External interfaces	<i>The system uses a trained Linear Regression model and local SQLite database.</i>	Medium

4	Transactions processing	Displays the predicted HDI score, development category, and data visualization insights.	High
5	Reporting	Displays the predicted HDI score, development category, and data visualization insights.	High
6	Business rules, etc.	Classifies the predicted HDI as Very High, High, Medium, or Low Human Development.	High
7	Compliance to laws or regulations	The system avoids collecting sensitive personal data and follows basic data privacy practices.	Medium
8	Other	Provides Home, Predict HDI, Result, About Project, and EDA visualization features.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The application should process inputs and display predictions quickly.</i>	<i>Average response time < 2 seconds</i>

2	Scalability	<i>The system should support increased users and larger datasets after deployment.</i>	<i>Supports multiple concurrent prediction requests</i>
3	Security & Data Privacy	<i>User inputs should be validated and sensitive data not collected.</i>	<i>100% input validation and no sensitive data storage</i>
4	Reliability & Availability	<i>The application should generate consistent predictions without unexpected failures.</i>	<i>Successful prediction rate > 99%</i>
5	Usability & Accessibility	<i>The interface should be simple, responsive, and easy to understand.</i>	<i>Prediction completed in a few simple steps</i>
6	Other	<i>The project should be modular, maintainable, and extendable.</i>	<i>Organized folder structure and reusable modules</i>